

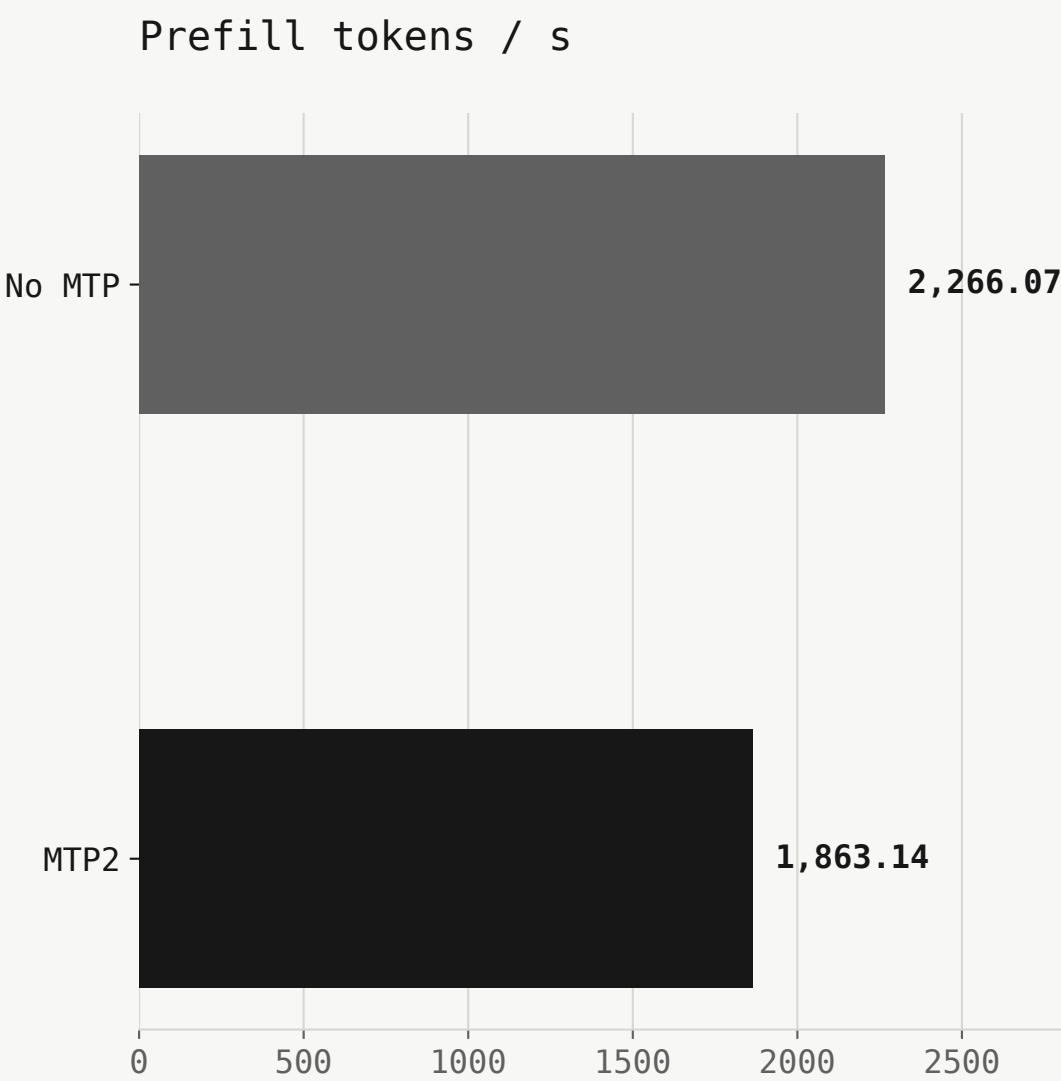
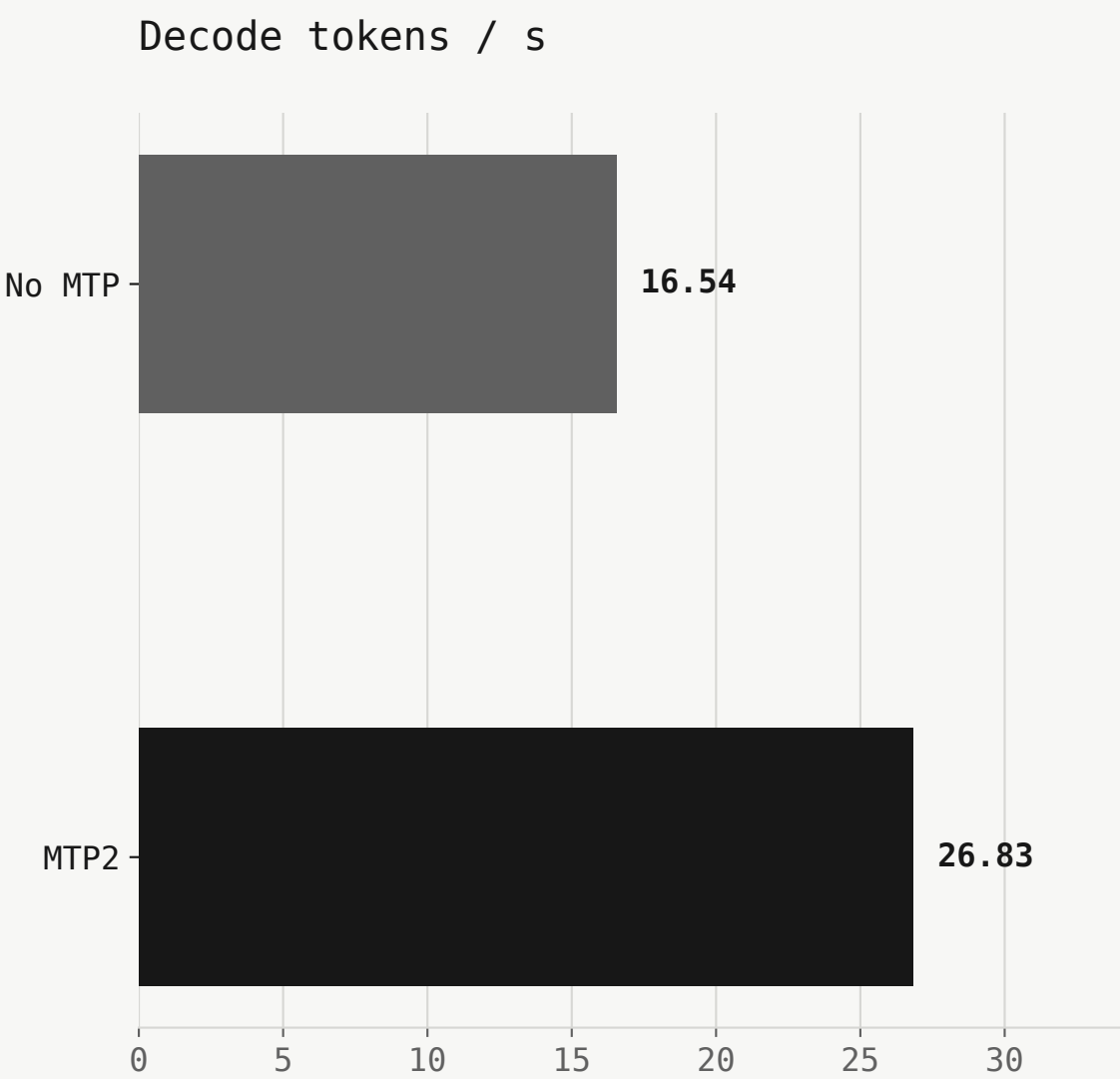
Qwen3.8 Flash Next : one GB10, 262K context

HISTORICAL CACHE-OFF SNAPSHOT / 2026-09-05 16:11 MDT / NVIDIA checkpoint / thinking enabled

Engine	vLLM 0.28.1rc1.dev442+g7fbd44cbe.d20260905 commit 7fbd44cbe0a90b9c8fd3a94a0f0401ac4b1bc719 + documented local fixes
Runtime	PyTorch 2.13.0 · FlashInfer 0.6.18 · Transformers 5.16.1 Python 3.12 · driver 580.173.02 / CUDA 13.0 · kernel 6.17.0-1031-nvidia
Hardware	Single GB10 / DGX Spark class UMA system · 20 ARM CPU threads 121.6 GiB OS-visible memory at test time · local NVMe checkpoint
Geometry	262,144 total context · 131,072 maximum shared reasoning + final output 8 GiB BF16 KV · batch 2,048 · one sequence · eager TP1 · native MTP2
Model structure	NVIDIA card: 125B main / 6B active + 51B n-gram embedding + 4B MTP 48 layers: 36 Gated DeltaNet + 12 sparse-attention layers 512 routed experts, 10 selected/token + shared expert · hidden size 2,560
Multimodal	27-layer vision encoder · one image/request · max_pixels 1,048,576 video disabled in this profile · qwen3 reasoning / qwen3_coder tools
Serving	nvidia/Qwen3.8-Flash-Next-NVFP4 · port 8092 medium thinking default; low / xhigh available · prefix cache OFF

MTP2 : faster decode, slower prefill

47,643 input tokens · identical request + sampling · medium thinking · 131,072 configured context



Decode +62.2% | Prefill -17.8%

One matched report, one run per setting; outputs differ (2,853 / 2,937 tokens).
Decode includes thinking, excludes the first token; phase timings from isolated server counters.

262K : populated context, measured answers

All rows thinking ON · BF16 KV 8 GiB · MTP2 · one sequence · bars show input tokens, not output speed

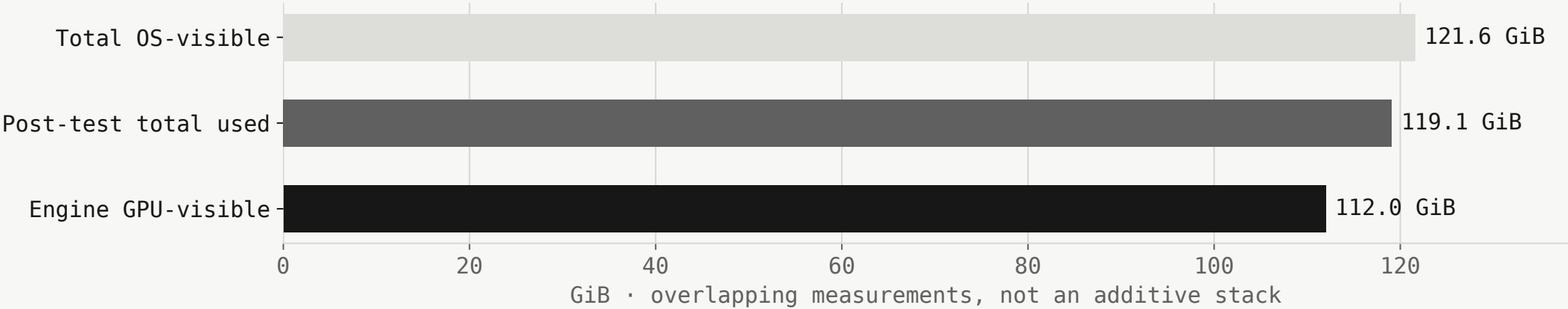


All four passed. Request times: 110.2s / 151.8s / 162.2s / 166.9s

Ledger: latest revisions + retracted correction + exact automatic tool arguments; follow-up passed.
Boundary: three dispersed needles. These are different tasks, not a controlled scaling curve.

Memory : the 8 GiB KV pool is only one part

Checkpoint tensors, allocated runtime memory and file-backed pages are different accounting domains



KV allocation

8 GiB BF16 pool → 288,802 advertised cache tokens
1.10× capacity ratio is not tested concurrency; max_num_seqs remains 1

PLE on disk

47.684 GiB original FP8, 128 shards, per-tensor scaling
CPU gathers selected rows from mmap; copied to GPU for dequantization

Pressure / limits

Post-test swap 20.6%; raw long-context MemAvailable observed about 2.9 GiB
No measured managed peak trace. Prefix reuse is disabled; cold prefill stays costly.

What changed : source fixes and integration repairs

Original NVIDIA checkpoint retained · no re-quantization · no community patch deployed

Existing PLE mmap	Preserved opt-in file-backed FP8 reader; requires eager TP1. Not upstream stock offload, asynchronous prefetch, or direct GPU NVMe reads.
MTP layer mapping	nvidia/mtp.py: map draft-local layer 0 to runtime layer 48. Preserve unrelated mappings; reject conflicts; use ModelOpt mixed config.
MTP block FP8	quantization/modelopt.py: route FP8_BLOCK_SCALES experts to generic block-FP8 MoE (128×128). Main NVFP4 expert dispatch remains intact.
PLE state stride	nvidia/ops/ple.py: upstream PR #55375 stride-aware state indexing. Speculation can supply non-contiguous indices; prefill/decode must honor strides.
Thinking / output	Keep thinking on; medium default. Server max_new_tokens=131072; OpenCode max_tokens=null avoids its smaller default and respects server headroom.
OpenCode sync / SSE	Preserve model options + variants and merge provider options. General, header and chunk timeouts all 1,200,000 ms (20 min).
Validation harness	Explicit default tmux socket for managed server; avoid inherited TMUX. Normalize 0.80 vs 0.8 in flag comparison; reuse passed stages after harness failure.
Checks / open work	16 selected engine tests + source pre-commit passed; 2 wrapper tests passed. MTP3, prefix-cache repair, full-output-length generation and long soak remain open.

Production : exact command line

Captured managed argv + relevant live environment · all paths anonymized

ENVIRONMENT

```
#!/usr/bin/env bash
set -euo pipefail
cd /home/user/src/vllm-upstream
export PATH=/home/user/venvs/vllm/bin:/usr/local/cuda/bin:$PATH
export CUDA_HOME=/usr/local/cuda
export HF_HOME=/home/user/vllm/hf_cache
export HF_HUB_OFFLINE=1
export MAX_JOBS=1
export NVCC_THREADS=1
export TMPDIR=/home/user/vllm/tmp
export TOKENIZERS_PARALLELISM=false
export VLLM_ALLOW_LONG_MAX_MODEL_LEN=1
export VLLM_CACHE_ROOT=/home/user/vllm/cache
export VLLM_QWEN4_PLE_MMAP=1
export VLLM_USE_DEEP_GEMM=0
export XDG_CACHE_HOME=/home/user/vllm/xdg_cache
```

Run environment setup first,
then the server command.

Existing managed server uses port 8092.
Do not start a second large engine.

All paths are anonymized.
Other home paths use /home/user.

SERVER COMMAND

```
exec /home/user/venvs/vllm/bin/python -m vllm.entrypoints.openai.api_server \
  --model /home/user/models/nvidia-Qwen3.8-Flash-Next-NVFP4 \
  --served-model-name nvidia/Qwen3.8-Flash-Next-NVFP4 \
  --host 0.0.0.0 \
  --port 8092 \
  --dtype bfloat16 \
  --kv-cache-dtype auto \
  --max-model-len 262144 \
  --max-num-batched-tokens 2048 \
  --max-num-seqs 1 \
  --gpu-memory-utilization 0.8 \
  --quantization modelopt \
  --kv-cache-memory-bytes 8G \
  --speculative-config '{"method":"mtp","num_speculative_tokens":2}' \
  --reasoning-parser qwen3 \
  --enable-auto-tool-choice \
  --tool-call-parser qwen3_coder \
  --limit-mm-per-prompt '{"image":1,"video":0}' \
  --skip-mm-profiling \
  --mm-processor-kwargs '{"max_pixels":1048576}' \
  --tensor-parallel-size 1 \
  --no-enable-prefix-caching \
  --enforce-eager \
  --load-format safetensors \
  --safetensors-load-strategy lazy \
  --generation-config vllm \
  --no-enable-flashinfer-autotune \
  --override-generation-config '{"max_new_tokens":131072}'
```

Validation : engine, wrapper, TUI, OpenCode

Promotion completed 2026-09-05 16:11:47 MDT · exact model ID and context verified at the endpoint

Direct engine	262K thinking features + 192K/250K ledgers + 260,834-token retrieval: PASS Separate raw launch with server-side 131,072 completion ceiling: PASS
Managed engine	Actual argv equals wrapper expectation; API model ID / max_model_len: PASS Thinking, retrieval, tools, tool results, vision and prose: PASS
TUI / OpenCode	Live UP row + executed warm action: PASS Real streaming tool execution + second request + final response: PASS
Production settings	Context 262,144 · output limit 131,072 · max_tokens null Thinking-only variants: low / medium / xhigh · 20-minute transport timeouts
Meaning of PASS	Exact checks above succeeded. Report completion does not certify every claim. Original agent corruption remains unexplained; later cap-truncation was distinct.
Not demonstrated	No MTP3 result, working prefix cache, concurrency >1, or extended soak. No 131,072-token generated answer; full output ceiling is a setting, not a load test.